

Algebra II with Trigonometry

Chapter 5.6

Study Guide (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4401272_alg-2trig-h-chp-5-6-note-docx.pdf](#)

Here is the information and study material you requested, based entirely on the provided document.

1. Summary of the Document

This document is a guided notes/worksheet packet for an Algebra 2 Trigonometry Honors class covering Chapter 5.6: The Law of Cosines. It focuses on solving oblique triangles specifically for Case 3 (Side-Angle-Side / SAS) and Case 4 (Side-Side-Side / SSS). The text introduces the Law of Cosines formulas for both finding sides and angles, defines navigational bearings, and introduces Heron's Formula for calculating the area of a triangle. It provides several empty example problems intended for in-class practice, ranging from basic triangle solving to real-world word problems involving kites, land value, and aircraft navigation.

2. Core Definitions, Theorems, and Formulas

- **The Law of Cosines (Verbal Definition):** The square of any side of a triangle is equal to the sum of the squares of the other two sides, minus twice the product of those two sides times the cosine of the included angle.
- **The Law of Cosines (Formulas for Finding Sides):**
 - $a^2 = b^2 + c^2 - 2bc \cos A$
 - $b^2 = a^2 + c^2 - 2ac \cos B$
 - $c^2 = a^2 + b^2 - 2ab \cos C$
- **The Law of Cosines (Formulas for Finding Angles):**

- $\cos A = \frac{b^2+c^2-a^2}{2bc}$
 - $\cos B = \frac{a^2+c^2-b^2}{2ac}$
 - $\cos C = \frac{a^2+b^2-c^2}{2ab}$
 - **Bearing (Navigation):** A direction given as an acute angle measured from due north or due south (e.g., $N 30^\circ E$ means starting North and rotating 30 degrees toward the East).
 - **Heron's Formula (Area of a Triangle):**
 - $Area = \sqrt{s(s-a)(s-b)(s-c)}$
 - **Semiperimeter (s):** Half the perimeter of the triangle, defined as $s = \frac{1}{2}(a+b+c)$.
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3. Worked Study Guide: Step-by-Step Concepts

Concept A: Using the Law of Cosines to Find a Side (SAS Case) When you know two sides of a triangle and the angle between them, you can find the side opposite that angle. 1. Identify your knowns (b , c , and $\angle A$) and your unknown (a). 2. Plug the values into the corresponding formula: $a^2 = b^2 + c^2 - 2bc \cos A$. 3. Calculate the right side of the equation. *Tutor tip: Be careful with order of operations! Compute $2bc \cos A$ entirely before subtracting it from the sum of the squared sides.* 4. Take the square root of the result to find side a .

Concept B: Using the Law of Cosines to Find an Angle (SSS Case)

When you know all three sides of a triangle, you can find any angle. 1. Identify the angle you want to find (e.g., $\angle A$) and note its opposite side (a). 2. Plug the sides into the angle formula: $\cos A = \frac{b^2+c^2-a^2}{2bc}$. 3. Evaluate the numerator and denominator separately, then divide. 4. Use the inverse cosine function ($\arccos$ or $\cos^{-1}$) on your calculator to find the measure of angle A .

Concept C: Finding the Area Using Heron's Formula If you only have the three side lengths (SSS) and need the area, Heron's Formula is the fastest route. 1. Find the semiperimeter (s) by adding all three sides and dividing by 2: $s = \frac{a+b+c}{2}$. 2. Subtract each individual side from the

semiperimeter: $(s - a)$, $(s - b)$, and $(s - c)$. 3. Multiply s by those three differences. 4. Take the square root of that entire product to get the Area.

4. Practice Problems with Answers

These problems are taken directly from the unworked examples in the provided document.

Problem 1 (From Example 1a) A triangle has two sides of length 25, and the included angle between them is 140° . Find the length of the opposite side x .

- **Answer:** $x \approx 46.98$
- **Solution Step:** $x^2 = 25^2 + 25^2 - 2(25)(25) \cos(140^\circ)$.

$x^2 = 625 + 625 - 1250(-0.766) = 1250 + 957.5 = 2207.5$. Taking the square root gives $x \approx 46.98$.

Problem 2 (From Example 8) A child is flying two kites at the same time, with 380 ft of line let out to one kite and 420 ft let out to the other. It is estimated that the angle between the two lines is 30° . Approximate the distance between the kites.

- **Answer:** ≈ 210.6 ft
- **Solution Step:** Let d be the distance.
 $d^2 = 380^2 + 420^2 - 2(380)(420) \cos(30^\circ)$.

$d^2 = 144,400 + 176,400 - 319,200(0.866) \approx 320,800 - 276,435.2 = 44,364.8$. Taking the square root gives ≈ 210.63 ft.

Problem 3 (From Example 4i) Find the area of the triangle whose sides have the given lengths: $a = 1$, $b = 2$, $c = 2$.

- **Answer:** ≈ 0.97 square units
- **Solution Step:** First find the semiperimeter, $s = \frac{1+2+2}{2} = 2.5$.

Using Heron's Formula:

$$\text{Area} = \sqrt{2.5(2.5 - 1)(2.5 - 2)(2.5 - 2)} = \sqrt{2.5(1.5)(0.5)(0.5)} = \sqrt{0.9375} \approx 0.97$$

Problem 4 (From Example 9) Land in downtown Columbia is valued at \$20 a square foot. What is the value of a triangular lot with sides of lengths 112, 148, and 190 ft?

- **Answer:** \$165,558.14 (or roughly \$165,560 depending on rounding rules)
- **Solution Step:** First, find $s = \frac{112+148+190}{2} = 225$.

Find Area

$= \sqrt{225(225 - 112)(225 - 148)(225 - 190)} = \sqrt{225(113)(77)(35)} = \sqrt{68,558.14}$
sq ft. Multiply by \$20: $8277.907 \times 20 = 165,558.14$.

Problem 5 (From Example 2ii) Solve triangle ABC given $a = 3$, $b = 4$, and $\angle C = 53^\circ$.

- **Answer:** $c \approx 3.25$, $\angle A \approx 47.6^\circ$, $\angle B \approx 79.4^\circ$.
- **Solution Step:**

1) Find c :

$$c^2 = 3^2 + 4^2 - 2(3)(4) \cos(53^\circ) \rightarrow 9 + 16 - 24(0.6018) \rightarrow 25 - 14.44 \approx 10.56$$

$$. c \approx 3.25. 2) \text{ Find } \angle A: \cos A = \frac{4^2 + 3.25^2 - 3^2}{2(4)(3.25)} = \frac{16 + 10.56 - 9}{26} \approx \frac{17.56}{26} \approx 0.675.$$

$$\arccos(0.675) \approx 47.6^\circ. 3) \text{ Find } \angle B: 180^\circ - 53^\circ - 47.6^\circ = 79.4^\circ.$$

5. Assumptions and Ambiguities Resolved

- **Missing Worked Solutions:** Because the document is a blank worksheet containing only the mathematical prompts without worked solutions, I had to assume standard algebraic procedures and apply the provided formulas myself to generate the Practice Problems and Worked Study Guide.
- **Rounding Conventions:** The document explicitly states "rounded to two decimals" for Example 5, but provides no rounding rules for the other problems. In the practice problems, I assumed standard math class conventions, rounding intermediate steps to 3 or 4 decimal places and final answers to 1 or 2 decimal places.

- **Example 2iv Ambiguity:** The top of the document explicitly states it covers "Case 3 (SAS) and Case 4 (SSS)." However, Example 2iv provides side a , $\angle B$, and $\angle C$. This is an Angle-Angle-Side (AAS) case, which requires the Law of Sines, not the Law of Cosines. I resolved this by noting that later in the document (Example 3), the instructions clarify to "Use either the Law of Sines or the Law of Cosines, as appropriate."
- **"Northeast" Ambiguity (Example 6):** The problem states a car traveled east and then "northeast". In navigational terms, exact northeast implies a precise 45° angle from due East. I assumed exact mathematical northeast (45°) to resolve the geometry required to solve the unworked problem.